

Corrosion and its control



Introduction

Corrosion occurs with all metals except the least active noble metals such as gold and platinum.

Eg: Rusting of iron and steel.

The corrosion is a slow process taking place mainly on the metal surfaces, but the losses incurred due to corrosion are of high order. The losses cannot be estimated by considering only the metal loss. The indirect losses are much higher. The fabrication cost and the cost incurred in preventing the corrosion are also have to be taken into consideration. The corrosion in metal objects like equipments, instruments, chemical plants, structures etc., may make them inefficient, ineffective, useless and their unobserved defects due to corrosion may cause their failure. In such cases, the indirect losses may be much higher than the direct losses.

When a structure like building or bridge collapses due to corrosion problems the loss also includes the loss of humn life and properties, and the cost of subsequent repair and altermate arrangement. In some forms of corrosion, the corrosion is quite invisible and seen only when an accident takes place.



Corrosion once thought a problem concerned with only metallurgists and chemists but now it is a field of interdisciplinary study as metals are universal in their use. There cannot be any field of material use which does not use metals. May it be a small metal object like pin or a huge mettalic structure like bridge, corrosion impairs their proper function and use. Therefore, corrosion has been correctly referred to as cancer to metals.

The knowledge of corrosion is therefore of great importance to the scientific community in general, and engineers in particular. It is the engineer who plans, designs and constructs the final metal product for use. In the process, due importance need to be given to the corrosion aspect. This can be successfully done only if he knows the mechanism of corrosion and the possible methods and mechanisms of control.

Definition;

Corrosion is defined as “the destruction and consequent loss of metals through chemical or electrochemical attack by the environment”.

The process of corrosion in reality is the transformation of pure metal into its undesired metallic compounds. The life of the metal object gets shortened by the corrosion process. Corrosion can be fast or slow process depending upon the metal and the environment in which it is undergoing corrosion.

Eg: 1. Rusting of iron- a reddish brown scale formation on iron and steel objects. It is due to the formation of hydrated ferric oxide.

2. Green scales formed on copper vessels. It is due to the formation of basic cupric carbonate. $\text{CuCO}_3 + \text{Cu(OH)}_2$.

Corrosion of metal occurs either by direct chemical attack or by electrochemical attack on the metal by the corrosive environment.

Direct chemical attack takes place generally in the absence of moisture. This type of corrosion is known as dry corrosion in which the constituents of the corrosion medium directly combine with the metal due to their chemical affinity to the metal.

Most of the corrosion cases are electrochemical in nature, taking place by the electrochemical attack on the metal in the presence of moisture or a conducting medium. Such corrosion is known as wet corrosion.

Direct chemical corrosion or Dry corrosion

Direct chemical corrosion occurs due to the direct chemical reaction between the metal and the gases present in the corrosion environment. This type of corrosion is generally observed in the absence of moisture or a conducting electrolyte medium, and therefore known as **dry corrosion**.

Direct oxidation of metals and alloys on exposure to oxygen in air is a common example of chemical corrosion. Apart from oxygen, many corrosive gases such as HCl, H₂S, SO₂, Cl₂, F₂ etc., also react with the metal and alloy surfaces when they come in contact with them.

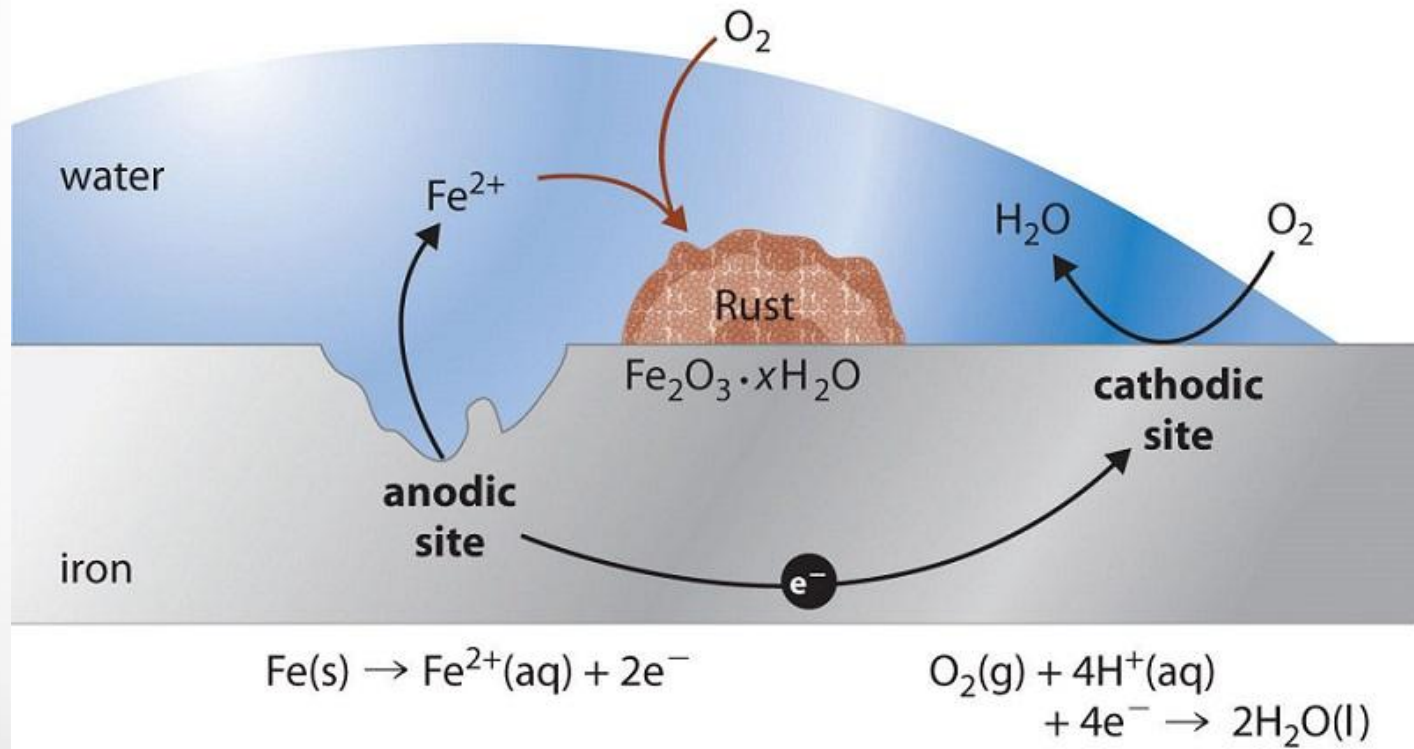
Two major factors deciding the rate of corrosion are the chemical affinity between the metal and the gas, and the nature of the corrosion product formed.

When a metal is exposed to a gas, it adsorbs the gas and slowly reacts with it and forms the corrosion product. For example, in oxidation corrosion, metal oxide is formed. The corrosion product so formed is deposited on the metal surface. Rate of further corrosion depends largely on the nature of the deposited corrosion product; whether it acts as a protective film or not.

Electrochemical theory of corrosion

Most of the corrosion problems are best explained on the basis of electrochemical corrosion. These are often called wet corrosion since aqueous medium or moist air is required for corrosion to take place.

According to electrochemical theory, corrosion of metals takes place due to the formation of anodic and cathodic regions on the same metal surface or when two different metals are in contact with each other in presence of a conducting medium.



At the anodic region oxidation reaction takes place and the metal gets converted into its ions, liberating electrons. Consequently, metal undergoes corrosion at the anodic region. At the cathodic region, reduction reaction takes place. Since the metal cannot be reduced further, metal atoms at the cathodic region are unaffected by the cathodic reaction. Some constituents of the corrosion medium take part in the cathodic reaction.

The electrons liberated at the anodic region migrate to the cathodic region constituting **corrosion current**. The metal ions liberated at the anode and some anions formed at the cathode diffuse towards each other through the conducting medium and form a corrosion product between the anode and the cathode.

Corrosion reactions



When iron undergoes corrosion,



But the cathodic reactions are more complicated than anodic reaction as the constituents of the corrosion medium are involved in the reaction, and are dependent on the nature of the corrosion environment. The most common types of cathodic reactions are either Liberation of hydrogen or absorption of oxygen. Correspondingly the corrosion types are known as hydrogen type corrosion and oxygen type corrosion.

Cathodic reactions:

a) Liberation of hydrogen takes place in the absence of oxygen

- In acidic medium and in the absence of oxygen, the cathodic reaction is



- In neutral or alkaline medium and in the absence of oxygen, hydroxide ions are formed with simultaneous liberation of hydrogen.



b) Absorption of oxygen takes place in the presence of oxygen

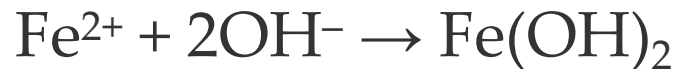
- In acidic medium and in the presence of oxygen



- In neutral or alkaline medium and in the presence of oxygen, hydroxide ions are formed



Corrosion of iron produces Fe^{2+} ions and OH^- ions at the anode and cathode sites respectively. These ions diffuse towards each other. Since smaller Fe^{2+} ions diffuse more rapidly than OH^- ions, their combination occurs more commonly near the cathodic region to produce insoluble $\text{Fe}(\text{OH})_2$. In an oxidizing environment it is oxidized to ferric oxide and yellow rust is hydrated ferric oxide.



In the presence of limited oxygen, ferrous hydroxide is converted into magnetic oxide of iron and is known as black rust.



Factors influencing corrosion rate

There are several factors which influence the rate of corrosion to a varying degree. Some of the important factors, affecting the corrosion rate are:

1. Nature of the metal:

The tendency of a metal to undergo corrosion is dependent on the nature of the metal. In general the metals with lower electrode potential values are more reactive than the metals with higher electrode potential values. The more reactive metals are most susceptible for corrosion.

Thus the tendency of a metal to undergo corrosion decreases with increase in electrode potential.

For example the so-called active metals like K, Na, Mg, Zn etc., with low electrode potential values are highly susceptible for corrosion. The noble metals such as silver, gold, platinum etc., with higher electrode potential values are less susceptible for corrosion. However, there are a few exceptions for this general trend as some metals show the property of passivity.

2. Difference in potential between anodic and cathodic region:

Larger the potential difference between the anodic region and cathodic region of the corrosion cell, higher is the corrosion rate. When potential difference is more, higher corrosion current is produced and the free energy decrease accompanying the process is higher and the corrosion rate is also higher. Therefore, when two different metals with large difference in the electrode potentials are in contact with each other, the more reactive metal undergoes corrosion very fast.

3. Nature of the corrosion product:

The nature of the corrosion product largely decides the rate of further corrosion. The corrosion product formed on the surface of the metal may or may not act as a protective film. If the corrosion product deposited is insoluble, stable, uniform and non porous it acts as a protective film preventing the further corrosion of metal. A thin, invisible, impervious continuous film formed on the surface acts as a barrier between the fresh metal surface and the corrosion environment.

On the other hand if the corrosion product is soluble, unstable, non-uniform and porous the corrosion continues. In such cases, the fresh metal surface is continuously exposed to the corrosion environment and corrosion of the metal surface takes place continuously.

In oxidizing environments metals like aluminium, chromium, titanium etc., are highly passive as their oxides as corrosion products forms protective films on the metal surface preventing further corrosion. Metals such as iron, zinc, magnesium etc., do not form any protective film and are highly susceptible for continuous corrosion, when exposed oxidizing environment. The protective films formed by the so-called passive metals are of such a self healing nature that, when ruptured they themselves get repaired on exposure to the corrosion environment.

4. The ratio of anodic to cathodic area:

The rate of corrosion is greatly influenced by the relative sizes of anodic and cathodic areas.

If a metal has a small anodic area and large cathodic area (i.e, the ratio of anodic to cathodic area is small), then the corrosion is more intensive and faster is the rate of corrosion at anodic region. As this ratio decreases, the corrosion rate increases rapidly. This is explained as follows.

At anode, oxidation takes place and electrons are liberated. At the cathode, these electrons are consumed. When the anode is smaller and cathode region is large all the electrons liberated at the anode, are rapidly consumed at the cathode region. This process makes the anodic reaction to take place at its maximum rate, thus increasing the corrosion rate.

If the cathode is smaller, the consumption of electrons will be slower and the corrosion reaction as a whole will be slower.

5. Temperature:

The rate of a chemical reaction, in general, increases with rise in temperature. Corrosion process is one such chemical reaction. Therefore, the rate of corrosion increases as the temperature increases.

Increase in temperature increases the conductance of the corrosion medium, which also contributes to the increase in corrosion rate. In the case of corrosion resistant passive metals, the rise in temperature decreases the passive range and thereby increases the corrosion rate.

6. pH:

In general, lower the pH of the corrosion medium, higher is the corrosion rate. However, some metals like Al, Zn etc., undergo fast corrosion in highly alkaline solution. The pH of the solutions also decides the type of cathodic reaction.

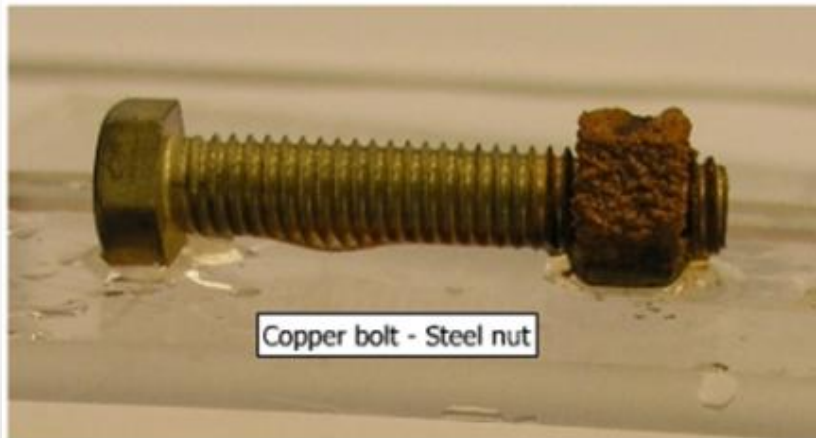


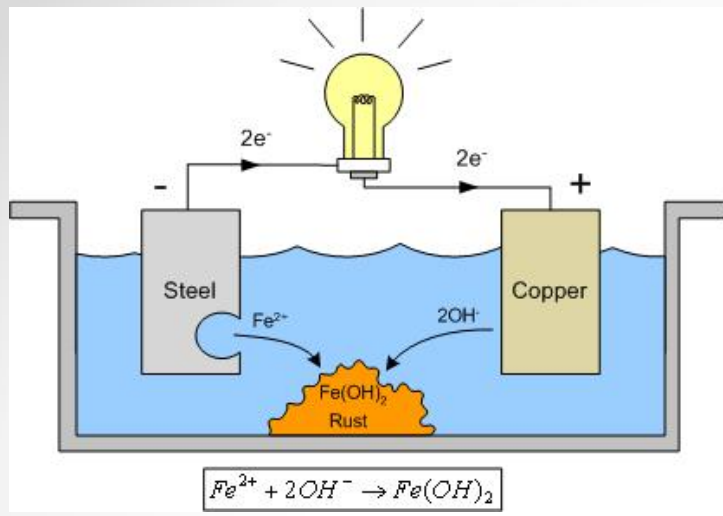
Types of corrosion

1. Differential metal corrosion/ Galvanic corrosion

Differential metal corrosion arises when two dissimilar metals are in contact with each other. Under this condition, a potential difference is set up resulting in a galvanic current. The metal higher in the electrochemical series is anodic to the one below it. For example, iron is anodic to copper, nickel and tin, but cathodic to zinc and magnesium. Therefore iron corrodes in contact with copper whereas zinc and magnesium corrode in presence of iron.

The rate of differential metal corrosion depends primarily on the difference in the potentials; higher the difference higher the rate of corrosion.

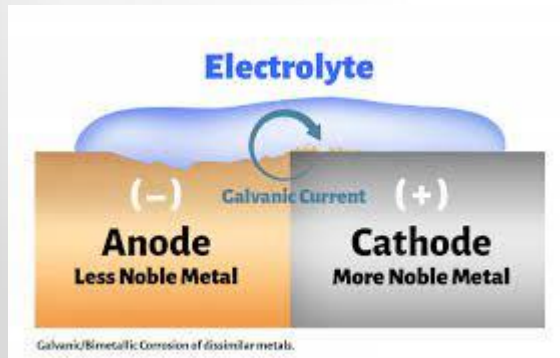




In this cell, iron and copper are in contact with an aerated solution of sodium chloride. A current flows between the two electrodes spontaneously and electrons flow from Fe to Cu indicating that Fe is the anode and Cu is the cathode.

The principle of differential metal corrosion occurring in a simple cell can be extended to a bimetallic couple.

If a piece of iron combined with copper metal is exposed to corrosive environment a galvanic current flows. Iron acts as anode in presence of copper. In such a bimetallic couple, iron undergoes corrosion and copper is unaffected. On the other hand in a bimetallic couple of iron and zinc, zinc gets corroded and iron is unaffected because zinc is more anodic.



When two metals are in contact with each other, a galvanic cell is set up and a current flows. The metal acting as anode undergoes corrosion and the metal acting as cathode is unaffected.

2. Differential aeration corrosion

Differential aeration corrosion occurs when a metal surface is exposed to differential air or oxygen concentrations. The part of the metal exposed to higher oxygen concentration acts as cathodic region and the part of the metal exposed to lower oxygen concentration acts as anodic region. Consequently the poorly oxygenated region undergoes corrosion.

At the anode (less O₂ concentration)



At the cathode (more O₂ concentration)



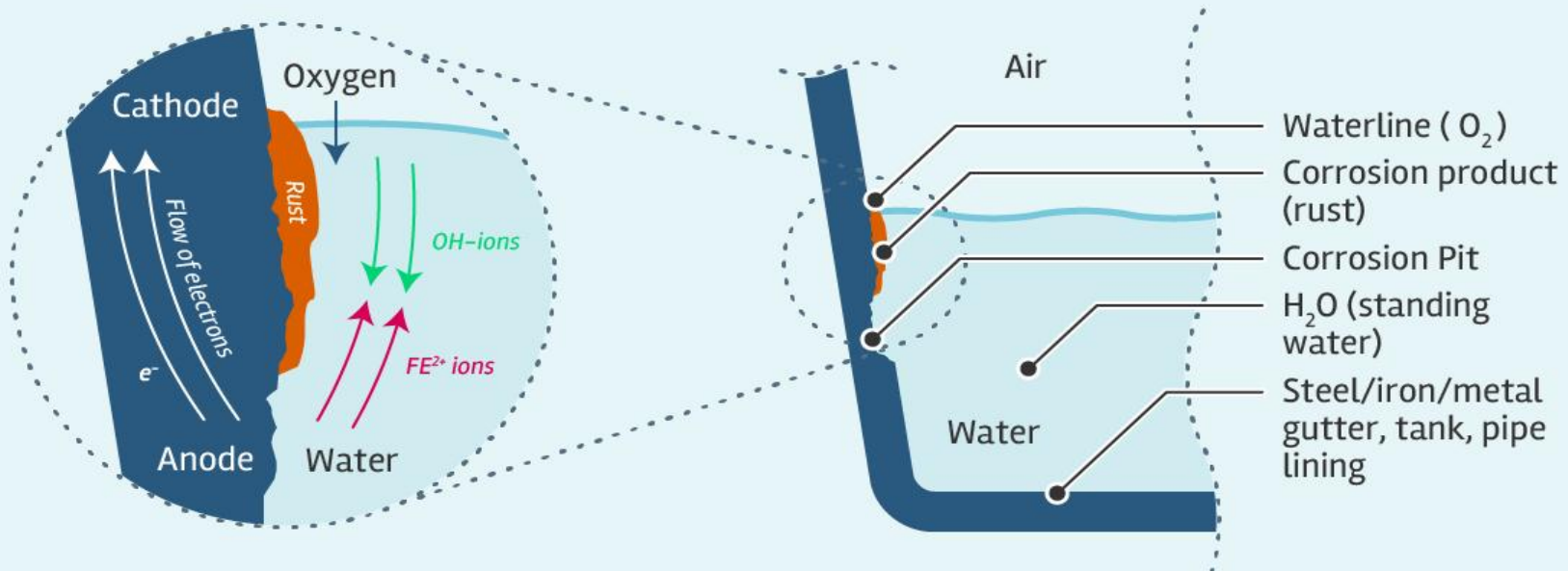
Corrosion of metals arising as a result of the formation of a oxygen concentration cell due to the uneven supply of air on the metal surface is known as differential aeration corrosion.

Eg:



Waterline corrosion

Differential Aeration Corrosion



The water line corrosion takes place due to the formation of differential oxygen concentration cell. The part of the metal below the water line is exposed only to dissolved oxygen while the part of the metal above the water is exposed to higher oxygen concentration of the atmosphere. Thus, part of the metal below the water acts as anode and undergoes corrosion and the part above the waterline is free from corrosion. A distinct brown line is formed just below the waterline due to the deposition of rust.

Waterline corrosion is common in water lifting pipes under water.

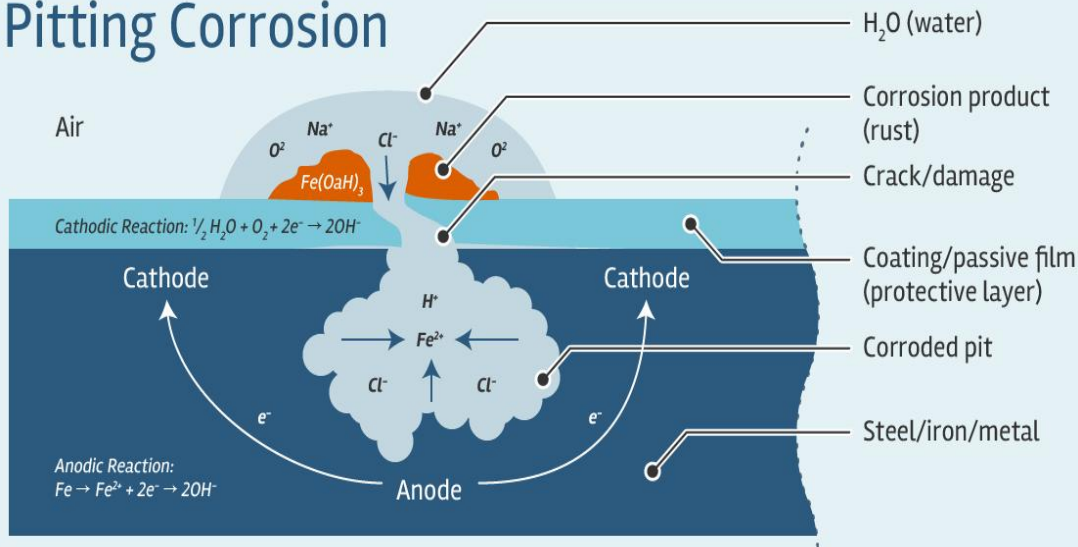
Though ocean going ships suffer from differential aeration corrosion, ships sunk under water for several years are free from corrosion. For the height of the ship sunk under water the difference in oxygen concentration between the top and bottom of the ship is negligibly small. Thus the ship is exposed to almost uniform concentration of oxygen and does not undergo differential aeration corrosion.

Pitting corrosion:

Pitting corrosion results when dust particles get deposited on a metal surface. The portion of the metal covered by the dust will be less aerated compared to the exposed surface. The covered portion thus becomes anodic with respect to the exposed surface.

In the presence of electrolytes and moisture, corrosion starts beneath the dust resulting in a pit. Once a pit is formed, the corrosion progresses rapidly. This is because of the formation of small anodic and large cathodic region. The latter induces the small anodic area to corrode faster by accepting electrons from the anodic area.

Pitting Corrosion



3. Stress corrosion

Stress corrosion occurs due to stress acting on the metal. The stress acting on the metal can be internal stress developed during the manufacture or fabrication of the articles, when the metals are subjected to mechanical operations such as pressing, hammering, bending, welding and reverting. The stress can also be external stress acting on the metal during service conditions.

The metal atoms under stress are always at higher energy levels as compared to the ones free from stress. The stressed part of the metal therefore becomes more reactive than the stress free part. As a result, a corrosion cell is formed with the stressed part acting as anode and the stress free part acting as cathode.



Under specific corrosion environment the stressed part undergoes corrosion, initiating the crack. In the presence of continuously acting stress and corrosion medium the crack propagates through the metal.

The corrosion medium is highly selective and specific in their action. For example, brass undergoes stress corrosion in the presence of ammonia and stainless steel in the presence of chloride ions and caustics.

Season cracking of brass and caustic embrittlement of steel are two classic cases of stress corrosion.

Caustic Embrittlement:

It is a form of stress corrosion that takes place in boilers operating at high pressure, at the stressed regions of the boiler. Boiler plate under stress becomes brittle by the action of caustic alkali which causes failure of the boiler. Caustic embrittlement occurs at stressed parts of the boiler such as cracks, rivets, bends and joints.

The boiler feed water invariably contains some sodium carbonate left behind during the softening process. At high pressure and temperature, the residual Na_2CO_3 undergoes hydrolysis producing sodium hydroxide.



The caustic water seeps through the minute cracks, between the rivets and joints by capillary action. Inside the cracks and crevices water gets evaporated leaving behind NaOH. The concentration of alkali gradually increases on these sites due to poor circulation of water. When the concentration of NaOH reaches a value of 10%, it attacks the metals at the stressed region, dissolving it in the form of sodium ferroate (Na_2FeO_2). This sodium ferroate undergoes hydrolysis, depositing magnetite (Fe_3O_4) by either of the following reactions.



OR



Sodium hydroxide is regenerated in the process. Thus the higher concentration of NaOH, once built up, continues to remain, maintaining the corrosion environment. The corrosion continues, developing cracks and making the metal brittle by the deposition of magnetite.

The corrosion cell can be represented as follows,

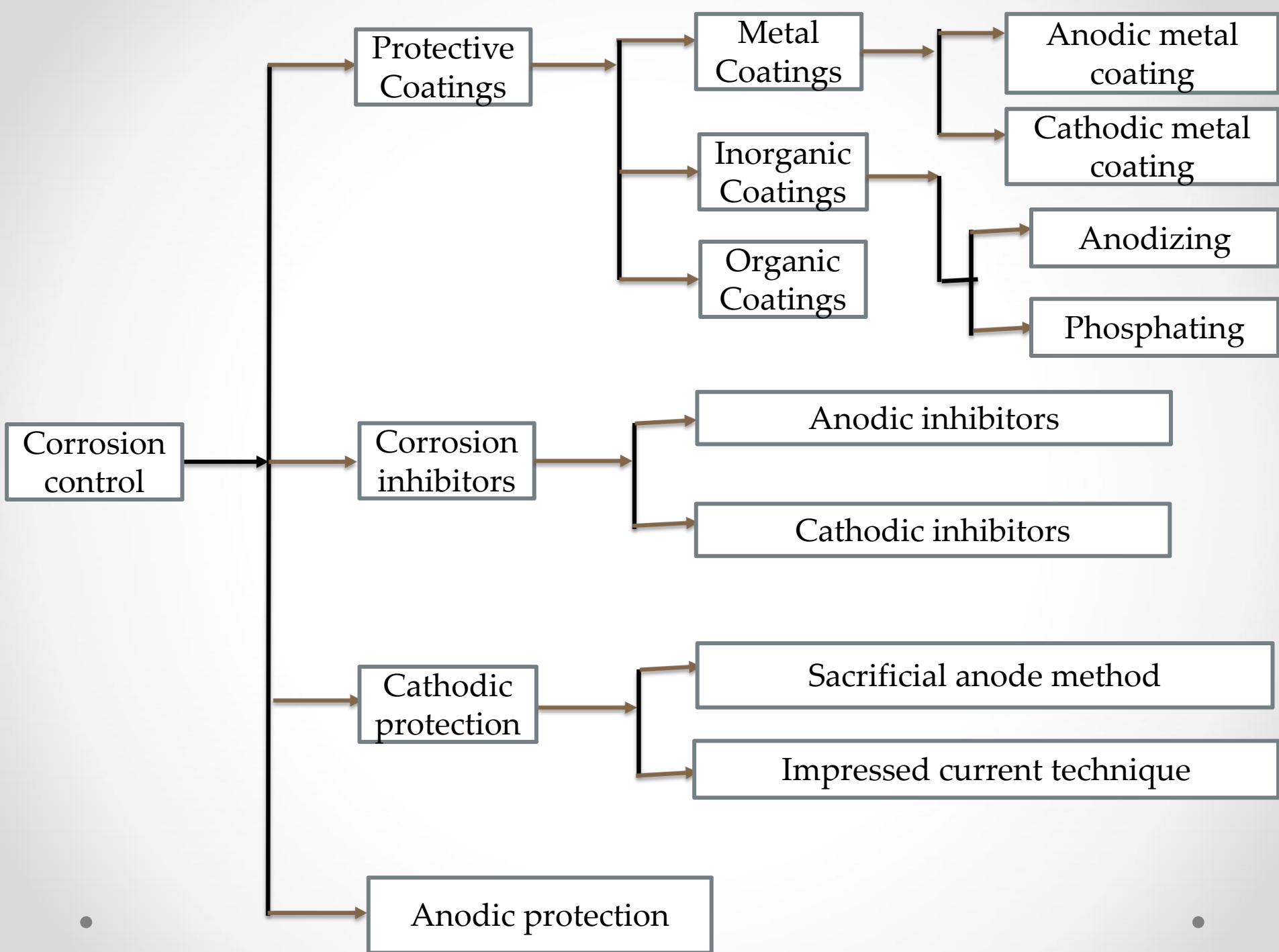
Fe under stress (Anode)/conc.NaOH/Dilute NaOH/Iron stress free (Cathode)

The iron in contact with concentrated alkali, being under stress, becomes anode while stress free iron surrounded by dilute alkali becomes cathode. Corrosion takes place at the stressed region. The continuation of caustic embrittlement of boiler parts often results in boiler failure or boiler explosion.

Prevention:

Caustic embrittlement can be prevented by the addition of compounds such as sodium sulfate, tannin, lignin, phosphate etc., which block the cracks or crevices, thereby preventing the infiltration of alkali.

CORROSION CONTROL



Metal coatings: Corrosion of metals can be controlled by isolating them from the environment. This can be brought about by covering the substrate or base metal with a layer of another metal. The process is referred to as metal coating.

1) Anodic metal coating:

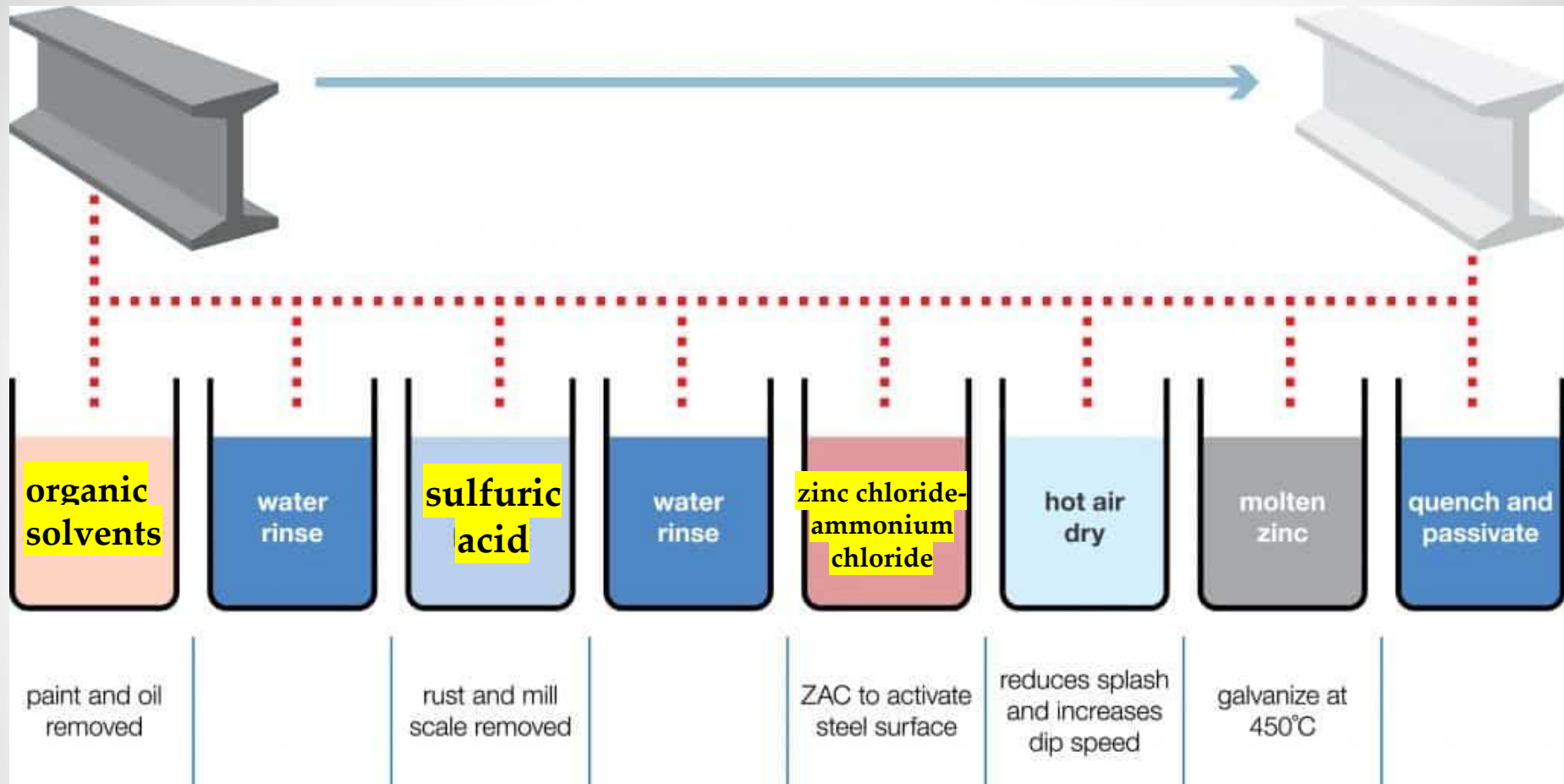
Anodic metal coatings are coatings which are anodic to base metal. Examples are aluminium, magnesium, zinc and cadmium coatings on iron. These metals are above iron in the galvanic series and undergo corrosion thereby protecting iron.

A characteristic feature of anodic coatings is that the base metal on which the coating is done will not get corroded even if the coating peels off. This is due to the formation of large anodic and small cathodic areas.

Galvanization: Coating of zinc on iron.

It involves the following steps:

- The metal surface is washed with organic solvents to remove grease or oil deposits.
- It is then treated with dilute sulfuric acid to remove rust and scale deposits.
- It is then washed with water and air dried.
- The clean and dry sheet is passed through zinc chloride-ammonium chloride flux. The flux helps the molten metal to adhere on the metal surface and also prevents the oxidation of zinc.
- It is dried and dipped in molten zinc at 430-470 °C.
- The excess zinc on the surface is removed by passing through a pair of hot rollers, which wipes out excess of zinc coating and produces a thin coating.



https://youtu.be/tCFCT_SP6so

Galvanization is used extensively to protect iron from corrosion in the form of roofing sheets, fencing wire, buckets, bolts, nuts, nails, screws, pipes, tubes etc. Galvanized steel reinforcements are used in building construction, where high degree of corrosion resistance is required.

Galvanized articles are however, not used for preparing and storing food stuffs, since zinc dissolves in dilute acids producing toxic zinc compounds.



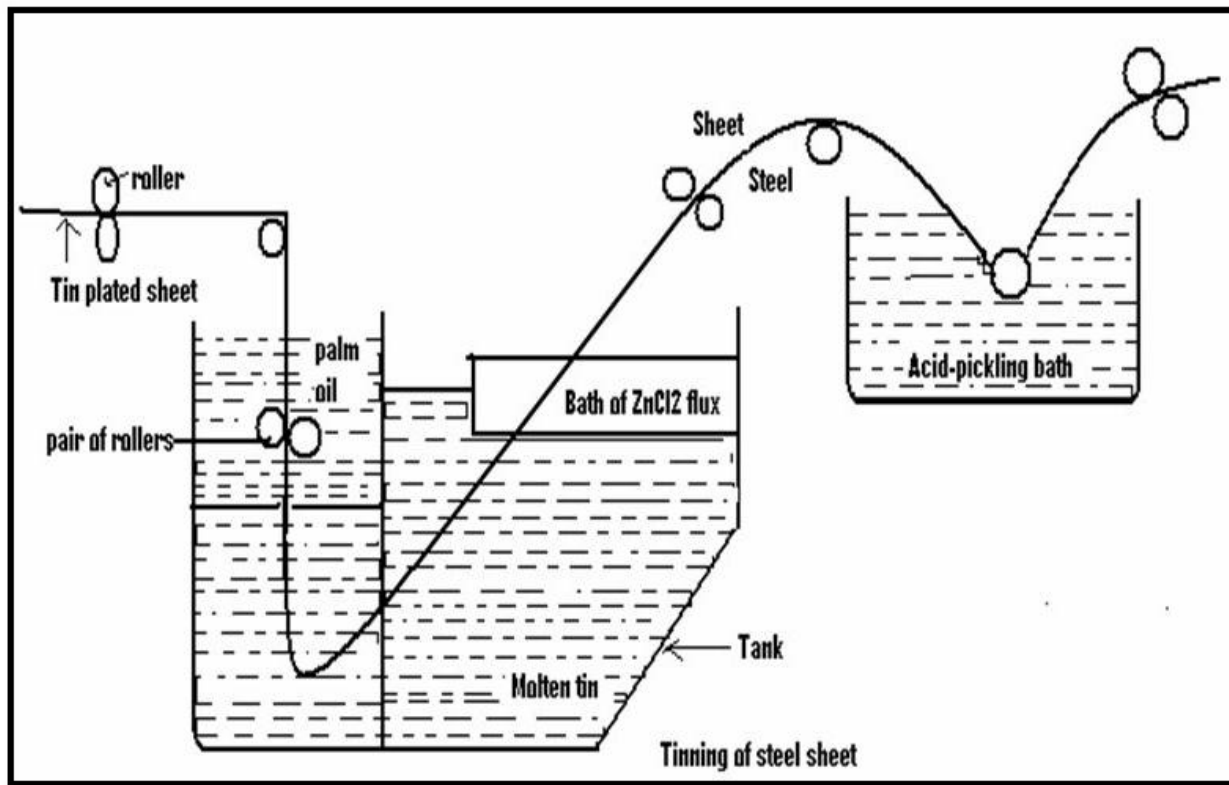
2) Cathodic metal coating:

Cathodic metal coatings are coatings which are cathodic to the base metal. Examples are **tin, nickel, chromium and copper on iron**. The coating is given in such a way that the applied coat completely covers the surface without leaving even pin holes. Otherwise rapid corrosion of the base metal takes place due to the formation of large cathodic and small anodic areas.

Tinning: Coating of tin on iron.

It involves the following steps:

- The metal surface is washed with organic solvents to remove grease or oil deposits.
- It is then treated with dilute sulfuric acid to remove rust and scale deposits.
- It is then washed with water and air dried.
- The clean and dry sheet is passed through zinc chloride-ammonium chloride flux. The flux helps the molten metal to adhere on the metal surface.
- It is dried and dipped in molten tin.
- The excess tin is removed by rolling.
- The coated tin is dipped in palm oil to prevent the oxidation of tin.



Tinning has the following advantages:

- Tin is nontoxic.
- Tinning gives a film which completely covers the surface and provides a corrosion resistant coating.

The largest single use of tin is in coating of steel for manufacturing containers used for storing food stuffs such as jam, instant food, milk products, pickles etc.

Inorganic coatings:

They are chemical conversion coatings. A surface layer of the metal is converted into a compound, by chemical or electrochemical reactions, which forms a barrier between the underlying metal surface and the corrosion environment.

The chemical conversion coatings are different from other types of coatings in the sense that, they are the integral part of the metal itself. This types of coatings are formed on the metal surface by chemical dip, spray or by electrolytic methods. In addition to the corrosion resistance, this type of coatings also provide increased electrical insulation and enhanced adherence for paints and other similar organic coatings.

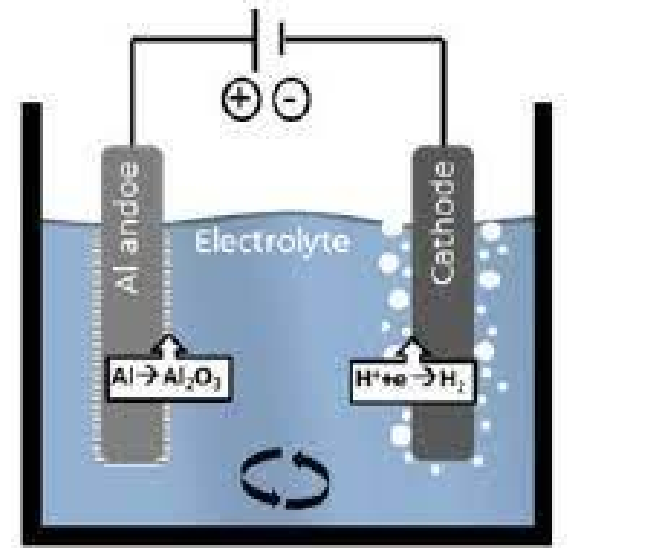
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1. Anodized Coating:

It is generally produced on non ferrous metals like Al, Zn, Mg and their alloys by anodic oxidation process, in which the base metal is made as anode in an electrolytic bath of suitable composition and by passing direct electric current. The most commonly used baths are chromic acid, sulfuric acid, phosphoric acid, oxalic acid, boric acid or their mixtures. Lead is generally used as cathode.

The anodic oxide film formed on Al in the bath of these materials tends to be porous and provides good adherence for paints and dyes. The strength and corrosion resistance of the anodized film can be increased by the so called sealing, which involves heating in boiling water or steam or metal salt solutions.

The treatment changes porous alumina at the surface of coating into its monohydrate ($\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$) which occupies more volume thereby the pores are sealed.





2. Phosphate Coating:

Phosphate coating is obtained generally on steel surface by converting the surface metal atoms into their phosphates by chemical or electrochemical reactions. Chemical phosphating includes dipping and spraying and electrochemical phosphating includes cathodic and anodic phosphating.

The phosphating bath contains three essential components:

1. Free phosphoric acid
2. A primary metal phosphate like Fe, Mn and Zn phosphates and
3. An accelerators such as nitrates, nitrites, chlorates, H_2O_2 etc.

The pH is usually within the range 1.8 and 3.2. Lower pH are favored for immersion deposits while spray processes often employ a high pH. The reaction includes dissolution of metal as ions, forming a phosphate with the bath solution which subsequently gets deposited on the surface of the metal.

Originally phosphate coating was applied only on iron and steel. But now it has been extended to Zn, Al and their alloys.

The most important application of phosphate coating is as an underlayer for paint finishes. Phosphating is a standard pretreatment before painting of automotive bodies and other steel sheet fabrications such as refrigerators and washing machines.

In the case of car bodies, electrophoretic priming usually follows the phosphating stage. Phosphating is widely applied for the protection of metal parts which cannot be painted or readily plated, such as nuts, bolts, moving parts of the machinery etc.

One of the important application is the phosphating of galvanized iron which is otherwise difficult to paint satisfactorily.



3. Organic Coating:

Coating of the metal surfaces with organic materials, particularly paints is the most widely used anticorrosion coatings.

The functions of organic coatings are of two folds:

- The coatings serve to keep out air and moisture from the metal surface or serve as a barrier between the metal surface and the corrosion environment.
- The pigments (red lead, zinc chromate etc.) or drying oils (linseed oil, wood oils etc.) present in the paint often exert an inhibitive action by electrochemical and other means.



The requirements of a good organic coatings are:

- It should adhere tenaciously to the metal surface and improve its physical appearance.
- The film formed should be continuous, uniform and impervious to air and water.
- It should be chemically inert to the corrosion environment.
- It should have reasonably long life.
- It should have proper application methods.

The performance of the paint depends to a larger extent on the application technique and even slight negligence at any stage may cause failure of the coatings.

The application of coating involves several steps.



Scrap off
Old Surface



Putty
Mixing



Apply
First Coat



Sanding the
Loose Particles



Final Coat of
Putty

Steps to Apply **WALL PUTTY**



Corrosion inhibitors

Corrosion inhibitors are the substances, which when added in small concentrations to a corrosive environment decreases the corrosion rate.

Corrosion inhibitors provides the protection against corrosion by retarding either the anodic or cathodic reactions.

They are mostly used to provide protection to systems in which, the corrosion environment is recirculated or confined for a longer period such as internal combustion engine, recirculating cooling water systems and pipelines for transporting oils, etc.

Corrosion inhibitors are classified into two types:

1. Anodic inhibitors
2. Cathodic inhibitors

Anodic Inhibitors

- At anodic region oxidation of metal takes place during the corrosion.
- For example, in the rusting of Iron, Fe^{2+} ions are formed at anodic site. If the formation of Fe^{2+} is prevented, the cathodic reaction stops and thereby the corrosion process is retarded.
- This is done by the addition of anions like chromate, tungstate, molybdate, phosphate etc. These anions combine with metal ions formed at anodic region forming the sparingly soluble respective salts, which gets deposited on the anodic sites forming protective films preventing further anodic reaction.

Examples:

- Chromates are used as inhibitors in rectifiers & cooling towers. Chromates and nitrites maintain the protective layer even in the absence of oxygen.
- Nitrites are used as inhibitors in antifreeze cooling water, internal surface of pipeline transporting gasoline and others

Cathodic Inhibitors

During the corrosion, at cathodic region, either liberation of hydrogen or absorption of oxygen takes place.

The two distinct approaches for inhibition of cathodic reactions is by preventing /retarding the liberation of hydrogen or by preventing/retarding the absorption of oxygen. This is achieved by –

1. Inhibition of hydrogen liberation :

This is done by two ways-

a) Preventing /retarding the diffusion of H^+ to cathode.

Ex: urea, thiourea, aliphatic amines, mercaptans etc.

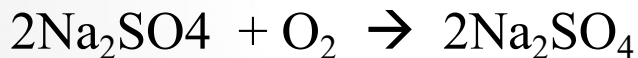
b) By increasing the hydrogen overvoltage on the metal surface.

Ex: oxides of arsenic, antimony or salts like sodium meta sulphite.

2. Inhibition of oxygen absorption

The absorption of oxygen can be reduced either by removing the oxygen from the corrosive media or by decreasing the diffusion rate of oxygen to cathode.

This is achieved by adding reducing agents or oxygen scavengers like hydrazine, sodium sulphite etc. to the corrosive medium.



It can also be achieved by adding salts like ZnSO_4 , MgSO_4 , NiSO_4 etc., into the corrosion medium



Cathodic Protection

Cathodic protection is a method of protecting a metal or alloy from corrosion by converting it completely into cathodic and no part of it is allowed to act as anode.

It is done by the following methods

1. Sacrificial anodic method
2. Impressed current method

Sacrificial anodic method

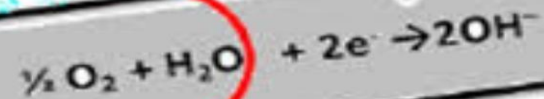
- In this method, the metal structure to be protected is converted into cathode by connecting it to another metal with lower electrode potential (active metals).
- Zn, Mg and Al are usually used as auxiliary anodes.
- These metals, being more active, act as anode and undergo corrosion protecting the metal structure from corrosion.
- Since the active metals are sacrificed to protect the metal structure, the method is known as sacrificial anode method.

Ex. A Mg block connected to a buried oil storage tank.

Sacrificial Anode
 $\text{Mg} \rightarrow \text{Mg}^{2+} + 2\text{e}^-$

2e^-

O_2



Protected Aluminum structure

$E_{\text{DCP, Mg}}^{\circ} = -1.55 \text{ V vs. SCE (in 0.1 wt\% NaCl solution)}$

$E_{\text{DCP, Al}}^{\circ} = -0.6 \text{ V vs. SCE (in 0.1 wt\% NaCl solution)}$

Advantages of this method:

- The method is simple
- Low maintenance cost
- Min maintenance cost
- Does not require power supply

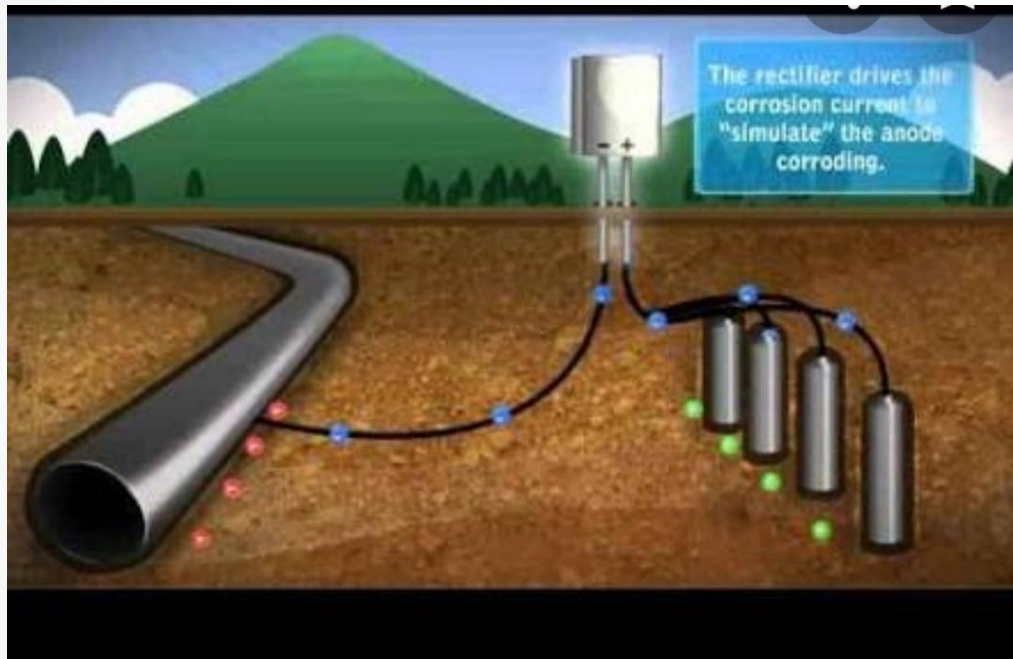
Disadvantages of this method:

- Involves recurring expenditure for replacement of consumed anodes

Impressed current method

- Impressed current technique is another method of providing cathodic protection is by applying direct current larger than the corrosion current.
- The protected metal is made cathodic by connecting it to the cathode of the external source of current. The anode of the source is connected to an inert electrode.
- The metal structure being cathode does not undergo corrosion. Anode being inert remains unaffected.
- Generally anode materials are graphite, Pt, Si, Fe are used in this method

Example for impressed current method



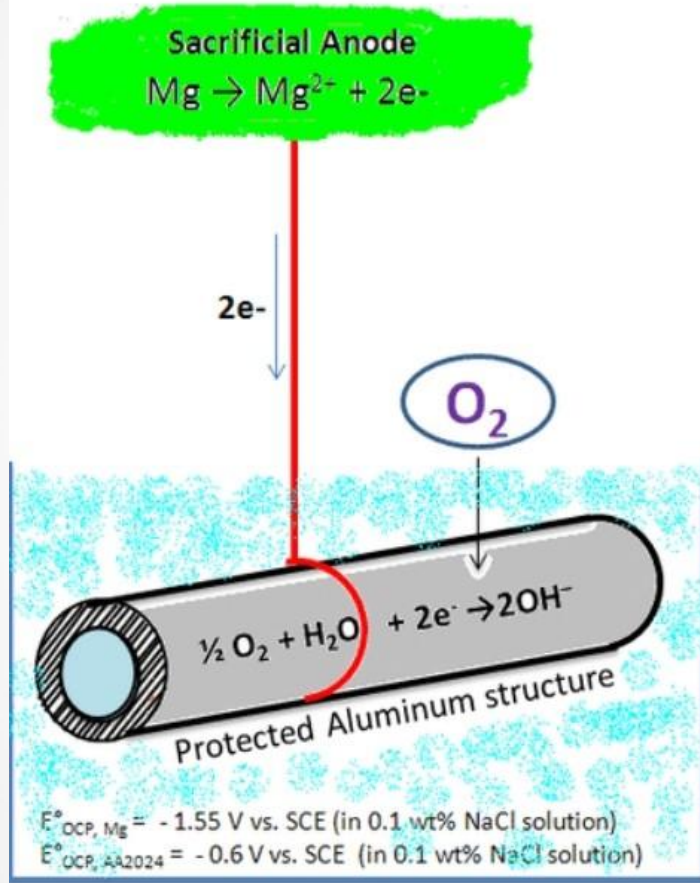
Advantages of this method:

- One installation can protect large area of metal
- Low maintenance cost

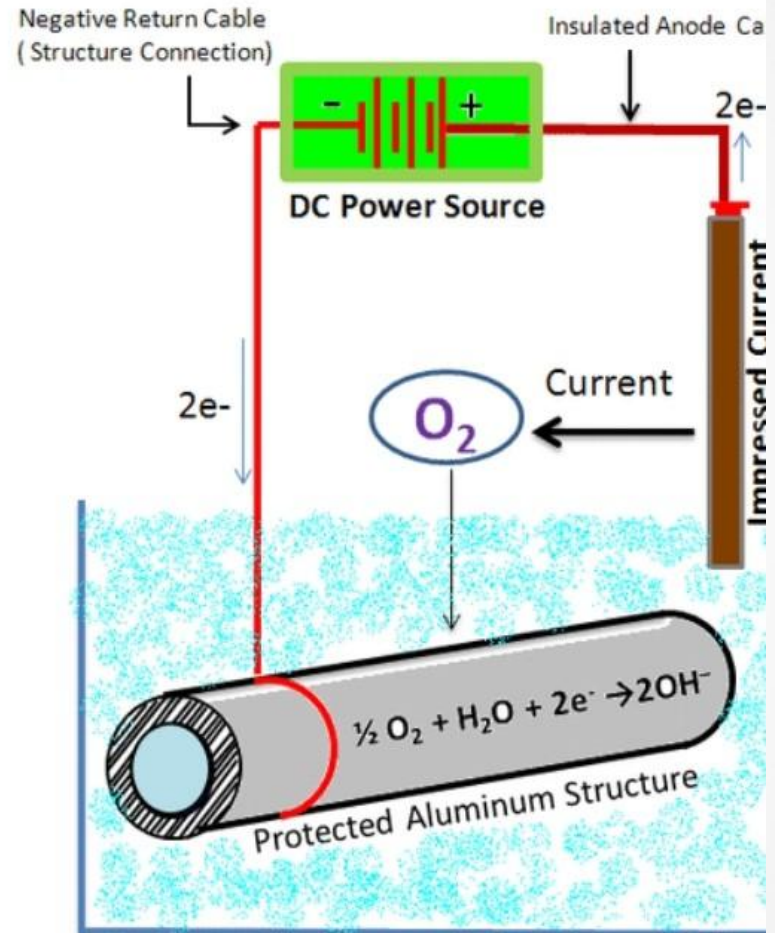
Disadvantages of this method:

- It needs high current for safe protection
- If the impressed current is not uniform on the entire surface of the protected structure, localized corrosion takes place

Sacrificial Anode CP Method



Impressed Current CP Method

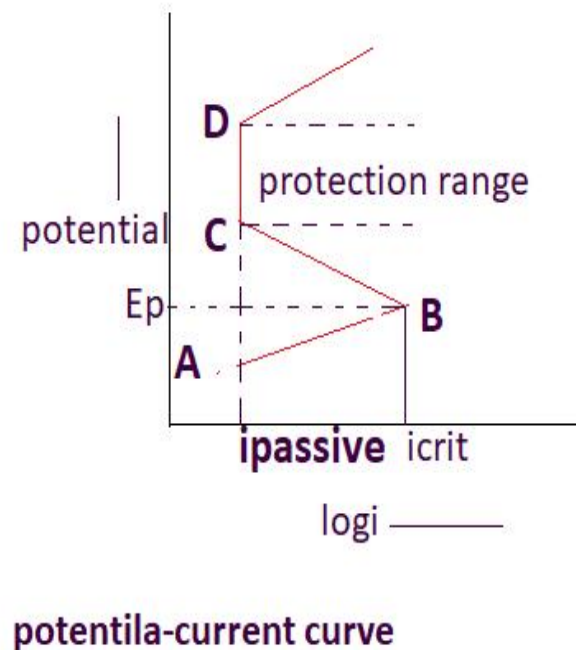


Cathodic Protection (CP) Methods

Anodic Protection

- Feasibility of anodic protection is firstly demonstrated and tested by Edeleanu in 1954 Corrosion control of metal structure by impressed anodic current.
- This method can be applied for *active-passive metal/alloys* such as Ti, Ni, Cr and their alloys only.
- Application of suitable anodic current makes them passive and decreases their rate of dissolution.
- The prevention of corrosion by impressed anodic current method is called *anodic protection*.

The potential required to protect the metal can be obtained from potential-current curve as shown in below figure.



- As the potential is increased, initially the current also increases(AB), indicating the dissolution of metal.
- This trend continuous until the current reaches a **critical value (i_{cri})** and passivated due to the development of oxide layers set in. This potential is called **passive potential(E_p)**.
- Above E_p , the current flow decrease and reaches a min value called the **passivated current i_p** .
- It is the min protective current density to maintain passivation.
- If the potential further increased, the metal remains unattacked till a particular potential reached(CD).
- This potential range, in which anodic protection is achieved is called **protection range**.
- Beyond the point D, any further increase in potential tends to increase the current due to the dissolution of metal. It is called trans passive region.

- The anodic protection to a structure is applied by using a device called potentiostat.
- For example, this method is used for protection of steel tank used for storing 96-98% sulphuric acid.

Advantages

- It is applicable in extremely corrosive environments.
- Low current demand

Disadvantages

- Anodic protection is restricted to the metals that show active-passive behaviour.
- Its initial installation cost is high.
- It can not reduce corrosion rate to zero unlike cathodic protection.

Thank You